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Paper 1 of 3 — CHM 310 Atomic and Molecular Structure

Level 300 • First Semester • PastQuestion

DEPARTMENT OF CHEMISTRY

COLLEGE OF SCIENCE

SCHOOL OF PETROLEUM RESOURCES, EFFURUN

EXAMINATION 2024/2025 ACADEMIC SESSION

AND MOLECULAR STRUCTURE AND

FEDERAL UNIVERSITY

FIRST SEMESTER

< COURSE TITLE: CHM 310 ATOMIC

AND MOLECULAR STRUCTURE

© INSTRUCTION: ANSWER ANY FOUR QUESTIONS TIME 2 HOURS 30 MINS

; tion for the wave mechanical model of an atom and

o ¥. (a) Derive the Schrödinger wave equation

discuss its application to the hydrogen atom (10mrks). —

(b) What is the significance of V and W in the Schrödinger wave equation? [2mrks]. _

(c) List two (2) significance and two (2) applications of the Schrödinger wave equation.

[5.5 mrks]

2. When an electron in y (f2) orbital (n2) to a

a hydrogen (11) atom falls from a higher energy

lower energy (E1) orbit (1), emission spectrum is obtained. , ;

me 1)

1) t * 'i werd ys } » ie j ay ih —_—_—_ —_—_ oo?

rd of the spectral line is given by Rydberg (a) nh

(a) Show that the wave number

[10mrks]

(b) Determine the Rydberg constant of the H-atom,

electron mass 9.109×10^{-31} kg; charge on electron 1.602×10^{-19} C, permittivity

medium 8.854×10^{-12} C²/N m²; Planck's constant 6.626×10^{-34} J s; Rydberg's constant

10973731.6 m⁻¹

A. (a) Write the possible terms or states of the atoms with the following configuration, and

identify which of the terms is the most stable in each case (i) $1s^2 2s^2 2p^4$ (ii)

$1s^2 2s^2 2p^6 3s^2 3p^4$ (iii) $1s^2 2s^2 2p^6 3s^2 3p^2$ (iv) $1s^2 2s^2 2p^6 3s^2 3p^4 4s^2 3d^2$ [10mrks]

(b) What is the dⁿ configuration and spin multiplicity of the ground state of V^{3+} and Cu^{+} ?

Fi (a) Determine the number of microstates in the following configuration (1) p^2 (ii) d^1 (iii) d^2 (iv) d^3 (v) d^4 (vi) d^5 (vii) d^6 (viii) d^7 (ix) d^8 (x) d^9 (xi) d^{10} (xii) d^{11} (xiii) d^{12} (xiv) d^{13} (xv) d^{14} (xvi) d^{15} (xvii) d^{16} (xviii) d^{17} (xix) d^{18} (xx) d^{19} (xxi) d^{20} (xxii) d^{21} (xxiii) d^{22} (xxiv) d^{23} (xxv) d^{24} (xxvi) d^{25} (xxvii) d^{26} (xxviii) d^{27} (xxix) d^{28} (xxx) d^{29} (xxxi) d^{30} (xxxii) d^{31} (xxxiii) d^{32} (xxxiv) d^{33} (xxxv) d^{34} (xxxvi) d^{35} (xxxvii) d^{36} (xxxviii) d^{37} (xxxix) d^{38} (xl) d^{39} (xli) d^{40} (xlii) d^{41} (xliii) d^{42} (xliv) d^{43} (xlv) d^{44} (xlvi) d^{45} (xlvii) d^{46} (xlviii) d^{47} (xlvix) d^{48} (l) d^{49} (li) d^{50} (lii) d^{51} (liii) d^{52} (liv) d^{53} (lv) d^{54} (lvi) d^{55} (lvii) d^{56} (lviii) d^{57} (lvix) d^{58} (l) d^{59} (li) d^{60} (lii) d^{61} (liii) d^{62} (liv) d^{63} (lv) d^{64} (lvi) d^{65} (lvii) d^{66} (lviii) d^{67} (lvix) d^{68} (l) d^{69} (li) d^{70} 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- A. (a) Identify which of these transitions is allowed or forbidden. (i) P_i
 (iii) $2P_4 \rightarrow D_3$, (iv) $7S_{u-P} \rightarrow 3P_3$ [12.5 mrks]
- (b) A metal complex was absorbed in the spectrometer at
 region did this adsorption occur? Calculate the energy of absorption. [5mrks]
- R_y in SI units. [mass of
 of the
 [7.5
-) of the spectral line emitted when the electron in
 $3 \rightarrow 2$, $n=3$ to $n=2$)
- D3 (ii) 'Se > Ps
- a wavelength of 650 nm. In which

Paper 2 of 3 — CHM 310 ATOMIC AND MOLECULAR STRUCTURE AND SYMMETRY 2024/2025

Level 300 • First Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
COLLEGE OF SCIENCE
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION
COURSE TITLE: CHM 310 ATOMIC AND MOLECULAR STRUCTURE AND SYMMETRY

INSTRUCTION: ANSWER ANY FOUR QUESTIONS TIME 2 HOURS 30 MINS

- ✓ 1. (a) Derive the Schrödinger wave equation for the wave mechanical model of an atom and discuss its application to the hydrogen atom [10mrks].
(b) What is the significance of Ψ and Ψ^2 in the Schrödinger wave equation? [2mrks].
(c) List two (2) significance and two (2) applications of the Schrödinger wave equation. [5.5 mrks]
2. When an electron in a hydrogen (H) atom falls from a higher energy (E_2) orbit (n_2) to a lower energy (E_1) orbit (n_1), H-emission spectrum is obtained.
(a) Show that the wave number $\bar{\nu}$ of the spectral line is given $\bar{\nu} = \frac{me^4}{8\epsilon_0^2 h^3 c} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$. [10mrks]
(b) Determine the Rydberg constant of the H-atom, R_H in SI units. [mass of electron = $9.108 \times 10^{-31} \text{ kg}$; charge on electron = $1.602 \times 10^{-19} \text{ C}$; permittivity of the medium = $8.850 \times 10^{-12} \text{ F}^1 \text{ C}^2 \text{ m}^{-1}$; Planck's constant = $6.626 \times 10^{-34} \text{ Js}$; $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$]. [7.5 mrks]
- ✓ 3. (a) Write the possible terms or states of the atoms with the following configuration, and identify which of the terms is the most stable in each case (i) $1s^2 2s^2 2p^1$ (ii) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$ (iii) $1s^2 2s^2 2p^6 3s^2 3p^2$ (iv) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$ [10mrks]
(b) What is the d^n configuration and spin multiplicity of the ground state of V^{3+} and Cu^{2+} ? [7.5mrks].
4. (a) Determine the number of microstates in the following configuration (i) p^2 (ii) p^1 (iii) d^1 (iii) d^5 [10 mrks].
(b) Calculate the frequency in (SI units) of the spectral line emitted when the electron in $n=3$ in H-atom de-excites to ground state. [7.5mrks]
5. (a) Identify which of these transitions is allowed or forbidden. (i) $^2P_{3/2} \rightarrow ^3D_{3/2}$ (ii) $^1S_0 \rightarrow ^1P_{1/2}$ (iii) $^2P_{3/2} \rightarrow ^3D_{3/2}$ (iv) $^2S_{1/2} \rightarrow ^2P_{1/2}$ (v) $^3D_0 \rightarrow ^3P_0$ [12.5 mrks]
(b) A metal complex was absorbed in the spectrometer at a wavelength of 650 nm. In which region did this adsorption occur? Calculate the energy of absorption. [5mrks]

.....END OF PAPER

Paper 3 of 3 — CHM 310 ATOMIC AND MOLECULAR STRUCTURE AND SYMMETRY 2024/2025

Level 300 • First Semester • PastQuestion

DEPARTMENT OF CHEMISTRY

COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

FIRST SEMESTER EXAMINATION 2024/2025 ACADEMIC SESSION

COURSE TITLE: CHM 310 ATOMIC AND MOLECULAR STRUCTURE AND

< SYMMETRY

© INSTRUCTION: ANSWER ANY FOUR QUESTIONS TIME 2 HOURS 30 MINS

X. (a) Derive the Schrödinger wave equation for the wave mechanical model of an atom and discuss its application to the hydrogen atom [10mrks].

(b) What is the significance of V and $\nabla^2\psi$ in the Schrödinger wave equation? [2mrks],

(c) List two (2) significance and two (2) applications of the Schrödinger wave equation. [5.5 mrks]

2. When an electron in a hydrogen (11) atom falls from a higher energy (n_2) orbit (n_2) to a lower energy (n_1) orbit (n_1), the emission spectrum is obtained.

a) $m_e \cdot 1.1$

(a) Show that the wave number ν of the spectral line is given by $\nu = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$.

Eg: $h \cdot c \cdot \nu = E_{n_2} - E_{n_1}$

[10mrks]

(b) Determine the Rydberg constant of the H-atom, R_H in SI units. [mass of electron = 9.108×10^{-31} kg; charge on electron = 1.602×10^{-19} C; permittivity of the medium = 8.85×10^{-12} F/m; Planck's constant = 6.626×10^{-34} Js]. [7.5 mrks]

A. (a) Write the possible terms or states of the atoms with the following configuration. and identify which of the terms is the most stable in each case (i) $1s^2 2s^2 2p^2$ (ii)

$1s^2 2s^2 2p^6 3s^2 3p^4 4s^1$ (iii) $1s^2 2s^2 2p^6 3s^2 3p^4 4s^1 3d^1$ (iv) $1s^2 2s^2 2p^6 3s^2 3p^4 4s^1 3d^2$ [10mrks]

(b) What is the d configuration and spin multiplicity of the ground state of V^{2+} and Cu^{2+} ? [7.5mrks].

(a) Determine the number of microstates in the following configuration (i) p^2 (ii) p^2 (ii) d^1 (ii) d^2 [10 mrks].

(b) Calculate the frequency in (SI units) of the spectral line emitted when the electron in $n=3$ in H-atom de-excites to ground state. [7.5mrks]

4. (a) Identify which of these transitions is allowed or forbidden. (i) $7P \rightarrow 7D$ (ii) $^1S_0 \rightarrow ^1P_1$ (iii) $7P_4 \rightarrow 3D_3$ (iv) $7S_0 \rightarrow 7P_1$, (v) $7D_0 \rightarrow ^1F_1$ [12.5 mrks]

(b) A metal complex was absorbed in the spectrometer at a wavelength of 650 nm. In which region did this adsorption occur? Calculate the energy of absorption. [5mrks]

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